

ATUL KUMAR

✉ Email | 🐙 Github | 🌐 Portfolio | 🔗 LinkedIn | 📄 Codechef

EDUCATION

Birla Institute of Technology, Mesra <i>Bachelor of Technology in Chemical Engineering</i>	2023 – Present Ranchi, India
Chinmaya Vidyalaya, Bokaro Steel City <i>Higher Secondary: Class: 11,12</i>	2021 – 2022 Bokaro, Jharkhand

EXPERIENCE

Web Development Intern <i>Zaalima Development (Remote)</i>	Apr 2026 – Present
- Developing a dashboard UI to manage and track 100+ AI-generated extensions, improving workflow visibility and user experience. - Implementing an “Edit Request” feature enabling real-time modification of generated code via prompts. - Building subscription-based feature gating for API-driven functionalities and actively deploying the subscription workflow. - Integrating code sanitization pipelines and conducting ongoing security audits to prevent malicious outputs in AI-generated code.	

PROJECTS

Automated Email Outreach System <i>Python, Pandas, SMTP</i> 🐙	Feb 2026
- Developed a Python CLI tool to automate email outreach to 100+ HR using structured CSV data. - Integrated secure Gmail SMTP with environment variable authentication to prevent credential exposure. - Built data validation and preprocessing pipeline to handle missing/invalid entries. - Implemented rate-limited batch delivery with error handling, reducing manual effort by 90%.	
Dental Caries Segmentation <i>YOLOv8, PyTorch, Computer Vision</i> 🐙	Jan 2026
- Developed a YOLOv8-based instance segmentation pipeline for detecting dental caries from 750+ radiographic images. - Trained on 768×768 images using AdamW optimizer with cosine learning-rate scheduling. - Applied augmentation techniques to increase effective dataset diversity by approximately 4×. - Achieved 22.6% mask recall using Precision, Recall, and mAP evaluation metrics.	
Market Regime Detection <i>Python, Pandas, NumPy, Scikit-learn</i> 🐙	Dec 2025
- Analyzed 7 years of NIFTY 50 data across 1,700+ trading sessions. - Engineered 10+ market indicators, including returns, volatility, and momentum features. - Applied clustering algorithms to identify 3 market regimes: Bull, Bear, and Sideways. - Discovered regime-specific return-volatility patterns for improved market interpretation.	

TECHNICAL SKILLS

Programming: Python, C, C++
Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Model Evaluation
Libraries: Scikit-learn, NumPy, Pandas, PyTorch
Databases & Tools: MySQL, DBMS, Power BI, MS Excel, Tableau
Technologies: Git, Docker, Jupyter Notebook
Web Development: HTML, CSS, JavaScript, WordPress
Cloud: AWS, Google Cloud

CERTIFICATIONS & JOB SIMULATIONS

Deloitte Data Analytics Job Simulation (Forage)
Data Analyst Bootcamp (Udemy)